Cross-Zone Load Balancing

Explore how cross-zone load balancing in AWS Elastic Load Balancers optimizes traffic distribution across Availability Zones. Understand its impact on fault tolerance, when to enable it, cost considerations, and practical steps to activate the feature using AWS CLI and Console. This lesson helps you build scalable, high-availability applications aligned with AWS best practices.

Ensuring traffic is properly balanced across resources is critical in a cloud environment designed for high availability. This lesson covers cross-zone load balancing in AWS Elastic Load Balancing (ELB) and its behavior across Availability Zones (AZs). We'll start by exploring how traffic is normally handled in AWS across zones, the default routing logic of ELBs, and what happens when cross-zone is not enabled. Then, we'll dive into how cross-zone load balancing works, identify situations where enabling it makes sense, and evaluate the associated costs, especially for Network Load Balancers (NLBs). We'll also learn about the performance gains, best practices, and potential pitfalls. Finally, we'll discuss how to enable cross-zone via the AWS Management Console and CLI, and close with a quiz to solidify our understanding.

Cross-zone load balancing in AWS ELB distributes traffic evenly across all healthy targets in all AZs, helping maintain performance and resiliency.

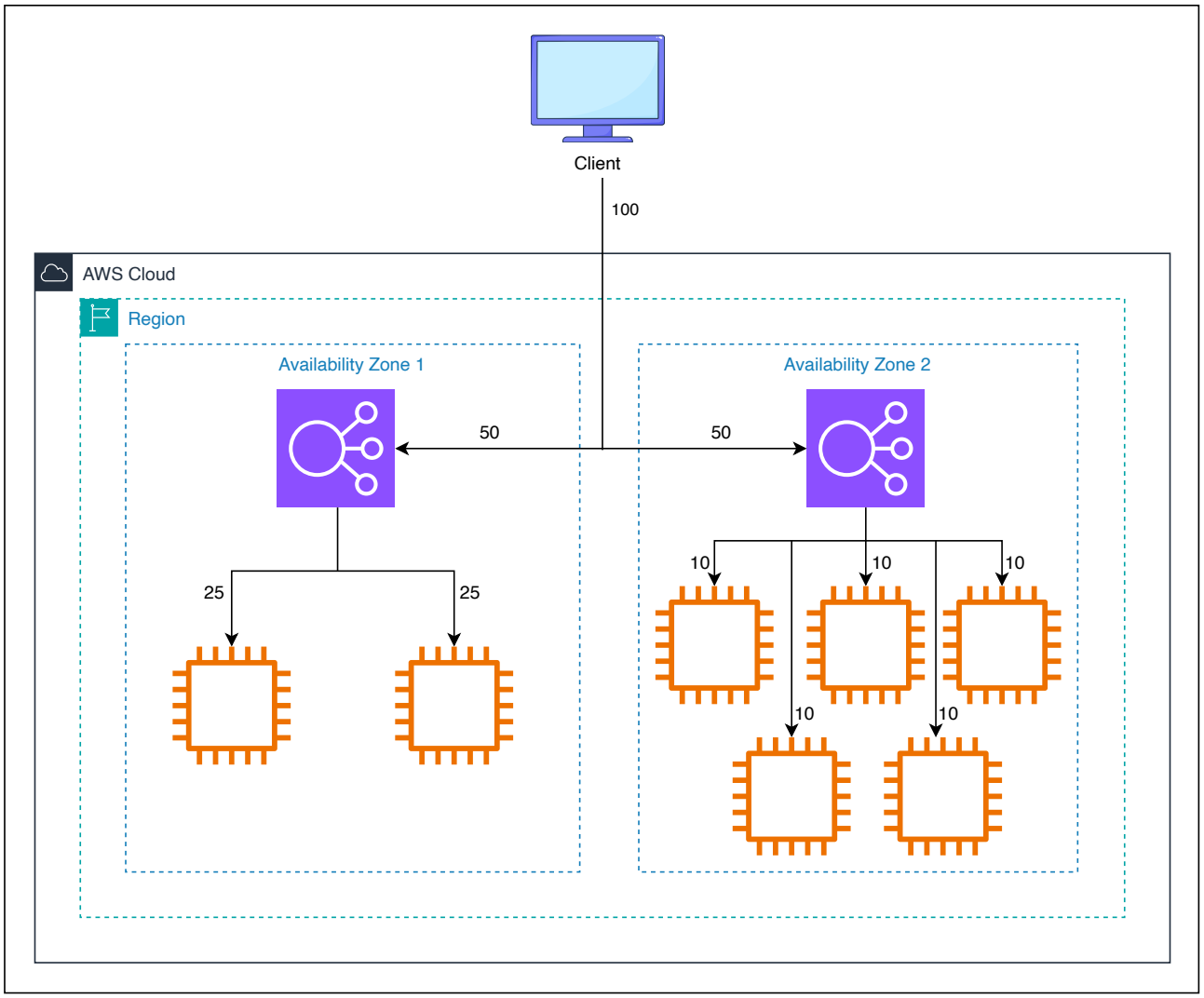
How cross-zone balancing works

Cross-zone load balancing allows ELB to route incoming requests to *any* h target in *any* AZ, regardless of the request's origin AZ. This ensures an even

Ask

distribution of traffic across all registered targets, optimizing availability and resource utilization.

Take an example below. In the first slide, load balancing is done without cross-zone balancing. The traffic is divided into two AZs without considering that AZ1 has only two healthy instances and AZ2 has five healthy instances. However, when we enable cross-zone balancing, the traffic is divided equally among all healthy instances.



1 / 2

Ask

Distributing traffic without cross-zone balancing

Cross-zone load balancing ensures even traffic distribution across all availability zones, which is crucial for maintaining balanced resource utilization. This leads to better fault tolerance, as a failure in one AZ won't overload another. For instance, in a high-traffic e-commerce application spread across three AZs, enabling cross-zone balancing can prevent bottlenecks by ensuring all servers share the load, improving overall system stability and user experience.

When to use cross-zone balancing

Cross-zone load balancing is ideal when our targets are unevenly distributed across AZs or when we want maximum fault tolerance and consistent traffic spread. It’s especially helpful during dynamic scaling events when some AZs may temporarily have fewer resources. However, it’s not always necessary. If we have evenly distributed targets and localized traffic patterns, the default AZ-level routing may suffice.

Cost differences we should know

Cross-zone load balancing can have cost implications, particularly with Network Load Balancers (NLBs). For ALBs, there is no additional cost. But with NLBs, cross-zone traffic incurs inter-AZ data transfer fees because requests may be routed from one AZ to another. These costs can accumulate if our workload sees high volumes of inter-AZ traffic.

Load Balancer Type	Cross-Zone Default	Cost Impact
Application (ALB)	Enabled (free)	None
Network (NLB)	Disabled (can enable)	Inter-AZ charges apply
Classic (CLB)	Enabled (free)	None

Ask

Best practices:

- Enabling cross-zone load balancing by default on ALBs since it's free and improves traffic distribution without downside.
- For NLBs, conduct a cost-benefit analysis before turning it on.

Enabling cross-zone load balancing using CLI

For the DVA exam, we need an overview of the steps via the console and CLI. We can modify the attributes of AWS CLI with the following command:

main.sh Shell-20

```
aws elbv2 modify-load-balancer-attributes \
  --load-balancer-arn <your-nlb-arn> \
  --attributes Key=load_balancing.cross_zone.enabled,Value=true
```

Modify attributes of load balancer

Suppose we have deployed an NLB across two AZs and want to ensure that traffic is shared evenly across both, even if one has fewer targets than the other.

- First, retrieve the ARN of the NLB using the `describe-load-balancers` command or from the AWS Management Console.
- Then, execute the `modify-load-balancer-attributes` command to the feature.
- Finally, verify that the change took effect by describing the NLB's attributes.

Ask

This will confirm that traffic is no longer bound by AZ constraints, helping us test resiliency and traffic flow behavior.

The expected output is structured as follows:

index.yaml YAML

```yaml
"Attributes": [
  {
    "Key": "load_balancing.cross_zone.enabled",
    "Value": "true"
  }
]
```

Expected output

Conclusion

Cross-zone load balancing is a powerful ELB feature that improves both availability and traffic efficiency. It's enabled by default for ALBs, while optional (and chargeable) for NLBs. Developers should weigh its benefits for even distribution and fault tolerance against potential cost trade-offs.

In this lesson, we looked at how ELBs handle cross-zone routing, why it matters when resources are unevenly distributed, and its broader impact on architecture, availability, and cost. We compared default behaviors across ALB, NLB, ar and practiced enabling cross-zone load balancing using the AWS CLI—bridging concepts with hands-on experience.

Ask

Understanding and managing ELB behavior is a key skill in the AWS Certified Developer—Associate (DVA-C02) exam, especially in the “Development with AWS Services” and “Deployment” domains. Knowing how and when to apply cross-zone balancing helps ensure scalable, resilient cloud applications that align with AWS best practices.

Ask